Research Papers on Agent Systems

OpenBot
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About this collection

These fifteen papers study distinct problems in agent systems. Each paper states its models, algorithms, assumptions, examples, and arguments directly. Each folder holds a complete Typst source file, its bibliography, and a PDF. Each paper uses charged-ieee 0.1.4.

The papers state methods, proofs under named assumptions, worked examples, and measured finite experiments. Each comparison method has explicit rules. The experiments include independent checks and ablations: runs that remove one rule to show why it is needed. Each paper separates these results from larger studies that have not been run. External research and standards are cited in each bibliography. The text states which methods are established and which result this paper adds.

Papers

01. Durable Agent Execution with Unknown External Outcomes
State, attempts, ownership, and unknown outside results.
Read PDF Typst source References

02. Authority Preservation Across Multi-Principal Agent Delegation
How identity and current rights bound each action.
Read PDF Typst source References

03. Layered Agent Authorization with Exact Grants, Cited Rules, and Escalation
How exact checks and model decisions share approval work.
Read PDF Typst source References

04. Completion Frontiers for Long-Lived Agent Tasks
How tasks progress and end while other work stays open.
Read PDF Typst source References

05. Evidence-Based Agent Memory from Self-Reported Learning Cues
How learning cues lead to supported and useful memory.
Read PDF Typst source References

06. Source-Constrained Information Flow for Shared Agent Memory
How source access limits derived knowledge.
Read PDF Typst source References

07. Dependency-Preserving Context Compaction for Tool-Using Agents
How context cuts keep tool and approval links whole.
Read PDF Typst source References

08. Replayable Tool Programs with Human Approval Boundaries
How a generated program pauses for approval and runs again.
Read PDF Typst source References

09. Secret References and Controlled Egress for Agent-Written Programs
How a program uses a secret without reading its value.
Read PDF Typst source References

10. Publisher-Grounded Discovery for an Open Agent Tool Web
How publisher evidence supports discovery and withdrawal.
Read PDF Typst source References

11. State-Based Conformance Testing for Agent Tool Protocols
How generated traces can find faults between implementations.
Read PDF Typst source References

12. Hierarchical Agent Budgets Under Uncertain Model Usage
How shared limits and delayed receipts guide admission.
Read PDF Typst source References

13. History Consistency for Interruptible Voice Agents
How output, playback, and committed history stay distinct.
Read PDF Typst source References

14. Capability-Aware Portability of Agent Tasks Across Compute Environments
How task needs match different execution environments.
Read PDF Typst source References

15. Preserving Execution Constraints in a Composable Agent Runtime
How ordered plugins preserve checks and resource ownership.
Read PDF Typst source References

Build

Install Typst and the TeX Gyre Termes and TeX Gyre Cursor fonts. From the project root, run:

```
devenv shell -- bash papers/build.sh
```

The build script accepts Typst options such as `--font-path`. Set `PAPERS_TYPST` to use another Typst binary. Each paper can also be built on its own by compiling its `main.typ` file. The import pins the template version. The PDFs in this collection are checked with Typst 0.15.1.

Reproduce the finite experiments

From the collection checkout, run:

```
devenv shell -- python3 papers/experiments/run.py \
  --seeds 30 --output papers/experiments/results.json
```

The runner uses Python's standard library. The saved results record the input domains, seeds, checks, observations, and limits for each paper. Source hashes cover the manuscripts, references, and experiment code. A run fails if these files change while it runs. Repeated runs should give the same model results; elapsed time and creation time can differ.

A check is one assertion, not an independent study or a user task. Exhaustive results cover only the named finite domain. Synthetic cases do not measure model quality, real network cost, or user behavior. The larger studies in each paper remain separate work. The runner and its model modules are maintained parts of this collection.

Research boundaries

Paper 1 owns recovery and outside effects. Paper 2 owns rights across identities. Paper 3 owns uncertain rule decisions and approval work. Paper 4 owns task coordination and completion. Paper 5 owns memory quality, while Paper 6 owns memory access. Paper 7 owns short-term context size. Paper 8 owns generated program replay, and Paper 9 owns secret values. Paper 10 owns open discovery. Paper 11 owns conformance test methods. Papers 12 through 15 own budget admission, live history, compute fit, and runtime composition, in that order.